

Building Water Efficiency Into Green Homes

by Justin Dunning

How much is 25 million gallons of water? Enough to fill 200 million 16-oz bottles. Enough bottles to keep you busy for the next 60 years, if you decide to count them down in “99 bottles of beer on the wall” fashion. Twenty-five million gallons is also the combined annual water savings of the more than 1,000 certified California Green Builder homes completed since 2005.

California Green Builder (CGB) is a voluntary program that provides a workable framework for production builders to build resource-efficient homes. Since 2005, over 3,000 homes have been enrolled in the program, and over 1,000 of those homes have been built. All CGB homes are built to be 15% more energy efficient than homes that just meet the standards set by Title 24—California’s energy code, one of the strictest in the nation. And all CGB homes are built to save a minimum of 20,000 gallons of water per year. In addition, CGB homes must use engineered-wood products from sustainable sources and must have an engineered HVAC design, with a MERV 6 filter for improved indoor air quality. And CGB builders must divert a minimum of 50% of construction waste from landfills. The energy and water savings for CGB houses are quantifiable; all the components of the program are verified by third-party inspections.

CGB was developed with the needs of many different stakeholders in mind. The result is a practical and user-friendly program that produces real results. The 1,054 homes that had been completed under the program as of March 8, 2007, will together save over 550 megawatt-hours of electricity per year, in addition to the 25 million



Jim Bayless, president of Treasure Homes, in front of his award-winning community, Fallen Leaf at Riverbend, in Sacramento, California. The homes feature reduced turf to qualify for the California Green Builder program.

gallons of water mentioned earlier—compared to Title 24-compliant homes. The ability to quantify savings helps to explain the program’s success; jurisdictions and homeowners alike want to know exactly what they are getting out of their green homes.

The one mandatory product in every CGB home is the use of a weather-based irrigation controller (WBIC). These units, which adjust the amount of water used for landscaping depending on the current climate, account for the lion’s share of the required reduction in water use. Other features commonly found in CGB homes are drip irrigation for nonturf plantings, parallel piping, hot water recirculation pumps, and high-efficiency toilets.

Drip irrigation for nonturf plantings not only saves water, but also increases the overall health of the landscaping. Drip irrigation uses less water than sprinklers do, since it waters only the

area around the roots of each plant. The water is also delivered more slowly; this keeps the soil moisture level more consistent and reduces evaporation caused by the buildup of surface water. By leaving the area between plants unwatered, drip irrigation discourages the growth of weeds. Finally, a drip irrigation system can be installed without trenching—another advantage over traditional sprinkler systems. A CGB home is credited with saving six gallons of water per square foot of landscaping per year.

Inside, too, water savings characterize a CGB home. In most homes, gallons go down the drain while residents wait for hot water to arrive from the water heater. CGB recognizes two solutions to this problem: parallel piping and hot water recirculation. Both solutions save water by reducing the volume of cold water that is wasted before hot water arrives at a fixture. A parallel piping system, also known as a manifold or

home run system, consists of a series of cross-linked polyethylene (PEX) pipes connected from a manifold to each fixture in the house. The PEX tubes are normally $\frac{3}{8}$ inch or $\frac{1}{2}$ inch in diameter, and they hold much less water than a traditional trunk-and-branch water supply system does.

system only at times when hot water is regularly in demand, such as before a morning shower. Installing either a parallel piping system or a recirc system in a home is credited with saving 7,000 gallons of water a year under CGB.

Home builders seeking additional savings can choose to install high-

bypasses, high-performance windows, radiant barriers, tankless water heaters, high-SEER air conditioning units, and ducts that have been well sealed and tested for leakiness. The performance of all CGB houses must be modeled using California Energy Commission-approved Micropas 7.3 residential-compliance software. A CGB compliance analyst conducts a plan check before approving a project's application to determine if a home will meet the requirements. In addition, CGB homes are inspected by certified HERS raters to verify that the correct features were installed as planned.

CGB's goal is to move California's home-building industry toward greater resource efficiency. Toward that goal, in March 2007, CGB, in partnership with the Imperial Irrigation District (IID), California's third largest utility, launched an incentive program offering builders up to \$1,700 per home for certifying their homes with CGB. This partnership model demonstrates how jurisdictions can use incentives to encourage the construction of better houses. Combined with the existing federal tax credit program, IID's incentive program greatly reduces the cost of energy- and water-efficient construction. By finding ways in which utilities and jurisdictions can benefit from green building, and by providing marketing support for green communities, CGB hopes to demonstrate that building green makes good business sense.



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For more information:

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VICTORY HOMES

Native plants, reduced turf, and drip irrigation help this Victory Homes community in Victorville, California, qualify for the California Green Builder program.

A hot water recirculation, or recirc, system saves the cold water that usually gets dumped down the drain while a person waits for hot water to arrive at a sink or shower. Instead of dumping this water, a recirc system pumps this cold water from the fixture back to the water heater. Some recirc systems pump continuously, so that hot water is constantly circulating through the house and instantly available at any fixture; however, these pumps use a significant amount of energy and heat is lost from the hot water as it circulates. Other recirc systems come with on-demand switches that pump cold water back to the water heater only when a hot water fixture is opened; the system is set to turn off as soon as the water at the fixture reaches a certain temperature (see "Benefits of Demand-Controlled Pumping," p. 48). In order to qualify for the CGB program, the pump must be fitted with either an on-demand switch or a timer. A timer is set to recirculate the

efficiency toilets. In the past, some poorly designed high-efficiency toilets that had insufficient flushing power have given these fixtures a problematic reputation. However, most of today's high-efficiency toilets have been redesigned, and published ratings of toilet performance are readily available (see "The Real High-Efficiency Toilets Have Arrived," p. 10). Dual-flush toilets also qualify as high efficiency. A dual-flush toilet has a regular and a reduced flush option; water savings are comparable to those of other high-efficiency toilets.

CGB offers builders a great deal of flexibility in meeting its energy efficiency requirements, just as it does with the water efficiency goals. The energy-efficient features found in a given house vary widely, depending on the builder and on the climate zone. Common energy-efficient features include higher R-value insulation carefully installed to avoid creating thermal